

NSU BSc CSE — 4-Year Course Plan

Fahim · Waived: MAT112, ENG102 · 134 credit-hour program · 12 terms (Fall / Spring / Summer each year) to graduate in 4 years at a lighter, ~9–13 credit/term pace

Year 1 — Fall (13 credits, 4 courses)

Course	Title	Prerequisite
CSE115 + 115L	Computing Concepts (+ Lab)	No prerequisite
MAT116	Pre-Calculus	No prerequisite
ENG103	Intermediate Composition	ENG102 waived
CHE101	General Chemistry	No prerequisite

Year 1 — Spring (12 credits, 4 courses)

Course	Title	Prerequisite
MAT120	Calculus & Analytic Geometry I	Needs MAT116
MAT125	Introduction to Linear Algebra	Needs MAT116
ENG105	Advanced Writing Skills	Needs ENG103
ENV107	Environmental Science	No prerequisite

Year 1 — Summer (10 credits, 3 courses)

Course	Title	Prerequisite
CSE135 + 135L	Fundamentals of Programming (+ Lab)	Needs CSE115, MAT120
PHY107 + 107L	General Physics I (+ Lab)	Needs MAT120
MAT130	Calculus & Analytic Geometry II	Needs MAT120

Year 2 — Fall (11 credits, 3 courses)

Course	Title	Prerequisite
CSE225 + 225L	Data Structures (+ Lab)	Needs CSE135
CSE243 + 243L	Electrical Circuits (+ Lab)	Needs MAT120
GED Elective 1	General Education	No prerequisite

Year 2 — Spring (13 credits, 4 courses)

Course	Title	Prerequisite
CSE173	Discrete Mathematics	Needs CSE135, CSE225
CSE253 + 253L	Electronics I (+ Lab)	Needs CSE243
MAT240	Calculus & Analytic Geometry III	Needs MAT130
GED Elective 2	General Education	No prerequisite

Year 2 — Summer (9 credits, 3 courses)

Course	Title	Prerequisite
PHY108 + 108L	General Physics II (+ Lab)	Needs MAT240, PHY107
MAT250	Calculus & Analytic Geometry IV	Needs MAT240
CSE231	Digital Logic	Needs CSE225, CSE173

Year 3 — Fall (12 credits, 4 courses)

Course	Title	Prerequisite
CSC273	Introduction to Theory of Computation	Needs CSE173, CSE225
CSE311	Database Systems	Needs CSE225
MAT361	Probability and Statistics	Needs MAT250
CSE232	Computer Organization and Design	Needs CSE231

Year 3 — Spring (12 credits, 4 courses)

Course	Title	Prerequisite
CSE327	Software Engineering	Needs CSE225
CSE323	Operating Systems Design	Needs CSE225, CSE232
CSE331	Microprocessors & Assembly	Needs CSE232
MAT350	Engineering Mathematics	Needs MAT250

Year 3 — Summer (9 credits, 3 courses)

Course	Title	Prerequisite
CSE338	Computer Networks	Needs CSE232, MAT361
CSE257	Numerical Methods	Needs CSE225, MAT125, MAT240
GED Elective 3	General Education	No prerequisite

Year 4 — Fall (11 credits, 4 courses)

Course	Title	Prerequisite
CSE326	Compiler Construction	Needs CSE232, CSC273
CSE373	Design and Analysis of Algorithms	Needs CSC273, MAT361
CSE413	Digital Electronics & Microprocessor Lab	Needs CSE331
CSE482	Internet and Web Technology	Needs CSE338

Year 4 — Spring (9 credits, 3 courses)

Course	Title	Prerequisite
Trail Elective 1	From chosen specialization trail	See trail note below
Trail Elective 2	From chosen specialization trail	See trail note below
Open Elective 1	Any approved open elective	No prerequisite

Year 4 — Summer (9 credits, 3 courses)

Course	Title	Prerequisite
Trail Elective 3	From chosen specialization trail	See trail note below
Open Elective 2	Any approved open elective	No prerequisite
CSE499	Project	Needs 100+ credits completed

Notes

- Every prerequisite above is satisfied by a prior term, never a concurrent one — confirmed against NSU's published CSE course descriptions.
- Total across all 12 terms is approximately 134 credits, matching NSU's published BSc CSE credit requirement (some third-party summaries cite 130 — catalogs get revised, so this is a strong model, not a guarantee).
- Trail electives (Year 4 Spring & Summer): pick one specialization trail and take 3 courses from it. Given the cybersecurity direction, the Computer Networks and Systems trail (CSE433, CSE438, CSE439, CSE463, CSE487) or the ICT trail (CSE437, CSE439, ETE426/427) line up best.
- This plan assumes attending all three terms every year (Fall, Spring, Summer) without a break. Skipping a summer term will push graduation past 4 years unless a later term's load is increased to compensate.
- Course availability varies by term and is not guaranteed by this plan — some required courses may only be offered in Fall or Spring, not Summer. Confirm each term's actual offerings and your live degree audit (RDS "Online Degree Analysis") with an academic advisor before registering.